

Om Prakash

Java Backend Engineer · Microservices · Spring Boot · REST API · RBAC · CI/CD

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<https://op2624685-sys.github.io/My-Portfolio>

PROFESSIONAL SUMMARY

Java Backend Engineer with production-grade experience delivering **high-throughput microservices** and enterprise **REST APIs** using **Java 8 / 17 / 21** and **Spring Boot**. Architected **Role-Based Access Control (RBAC)** systems, **JWT / OAuth2** authentication pipelines, and distributed transaction flows for real-world, deployed applications. Hands-on ownership across the full backend lifecycle — **PostgreSQL** schema design, **Redis** caching, **Kafka** event streams, **Grafana / Prometheus** observability, and **GitHub Actions** CI/CD pipelines on **Linux**. Practised **Unit Testing** and **Integration Testing** with **JUnit 5** and **Maven** across both production systems — with multi-role user management, payment integration, and data isolation.

TECHNICAL SKILLS

Languages: Java 8 / 17 / 21, Python, JavaScript

Backend: Spring Boot 4.1, Spring Security, Spring Data JPA, REST API, JWT, OAuth2, Kafka, WebSocket

Frontend: React.js, Next.js, Tailwind CSS, Shadcn UI, GSAP, Lenis, Lucide React, Axios, React Query, HTML5, CSS3

Databases: PostgreSQL, MySQL, MongoDB, Apache Cassandra

Caching: Redis

Monitoring: Grafana, Prometheus

Testing: JUnit 5, Unit Testing, Integration Testing, Maven (build & test lifecycle)

DevOps & CI/CD: GitHub Actions, Continuous Integration (CI), Continuous Deployment (CD), Docker, Git, Postman, Linux

Core Concepts: Microservices Architecture, RBAC, OAuth2, System Design, OOP, DSA, Payment Integration, Data Isolation

PROJECTS

Dadi Bulsara Ashihara Karate Federation — Management System · [GitHub](#) · [Live Demo](#)

Backend: Java 21, Spring Boot 4.1, Spring Security, JWT, OAuth2, REST API, PostgreSQL, Redis, FreePayment, Maven, JUnit 5, GitHub Actions

Frontend: Next.js, Shadcn UI, Tailwind CSS, Lenis, Lucide React, GSAP

- **Architected 5-tier RBAC system** (Normal User → Student / Teacher → SubAdmin → Admin) with scoped JWT claims, permission guards, and fully isolated dashboards per role.
- **Built payment-gated role-upgrade flow** using FreePayment: users pay at registration, SubAdmin verifies credentials, backend promotes role — with full audit trail for pending / rejected states.
- **Teacher Panel:** approve / reject student applications, manage attendance records, and maintain dojo-level data — all role-scoped with Spring Security method security.
- **Teacher public profile:** dojo address, Google Maps embed, enrolled students, belt certifications, achievements, quick Gmail contact — all served via optimised REST endpoints.
- **Student public profile:** belt level, join date, assigned teacher, dojo name (deep-linked), camp & tournament history, wins, and achievements — structured as separate DB entities with relational mapping.
- **SubAdmin dashboard:** state-scoped management of verifications, teacher / student counts, and payment records with PostgreSQL row-level data isolation.
- **Admin dashboard:** cross-state analytics — active teacher / student growth per state, revenue trends, global verification pipeline, and operational oversight.
- **Redis caching** on high-read endpoints (teacher / student listing pages) to reduce DB round-trips and improve response latency.
- **GitHub Actions CI/CD pipeline (Continuous Integration + Continuous Deployment):** automated **Maven** build → **JUnit** test suite → Docker image push → Linux server deploy on every merge to main.
- **Unit Testing & Integration Testing** with **JUnit 5** and **Maven** Surefire — service-layer unit tests and full REST endpoint integration tests with MockMvc and test containers.
- **Grafana + Prometheus** monitoring: real-time dashboards for API request rates, error rates, JVM heap, and DB connection pool health.

Advanced Hospital Management System

Backend: Java 8 / 17, Spring Boot, Spring Security, JWT, OAuth2, REST API, MySQL, Redis, Maven, JUnit 5

Frontend: React.js, React Query, GSAP, Axios, Vite

- **Designed RBAC with 4 actor roles** (Patient, Doctor, Admin, Super Admin) using Spring Security — each role with distinct API access scope and dashboard.

- **Built real-time appointment & facility booking engine** for Doctor consultations, X-ray, and Lab services with conflict-free slot scheduling.
- **Integrated online payment** with automated PDF invoice generation triggered post-payment via server-side PDF rendering.
- **Multi-branch architecture** with complete data isolation per branch — branch-level dashboards for distributed clinic management.
- **Google / Email / Phone OAuth2 authentication** via Spring Security with JWT session management.
- **Ambulance & Emergency module:** real-time dispatch tracking with live status updates.
- **Medicine inventory:** expiry-date tracking with automated low-stock and near-expiry alerts.
- **Super Admin cross-branch KPI dashboard:** staff attendance, occupancy rates, revenue, and operational metrics aggregated across all branches.
- **GitHub Actions CI/CD (Continuous Integration + Continuous Deployment):** **Maven** build → **JUnit** test suite → Dockerised Spring Boot image → automated Linux server deploy.
- **Unit Testing & Integration Testing** with **JUnit 5** and **Maven** Surefire — covering service logic, repository layer, and REST controllers via MockMvc.
- **Grafana + Prometheus observability:** API latency, error rate, and JVM metrics dashboards with alerting.

EDUCATION

Bachelor of Science (B.Sc.) – Botany

Thakur Prasad Singh College, Patna · Aug 2024 – Aug 2028 (Expected)

LANGUAGES

Hindi (Native) · English (Fluent) · Japanese JLPT-N4 (Intermediate)